

Household Wattch: Exploring Opportunities for Surveillance and Consent through Families' Household Energy Use Data

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Household energy use data may contain sensitive inferences into family life, yet its potential for surveillance is imperfectly understood. To explore this space, we developed Household Wattch, a speculative eco-feedback 'prototype' that profiles households according to their energy use data. Evaluated by 16 participants from Australian households engaged in an 18-month energy use monitoring trial, Household Wattch elicited users' perceptions and expectations about a near future where energy use data is a useful yet potentially sensitive commodity when analysed. We highlight challenges and opportunities for energy use data across three scales: (1) Within the household, (2) Beyond the household (e.g., sharing energy data with third parties) and (3) Post-household (e.g., what happens to energy data when a household re-configures or disbands). Findings suggest users may require support in understanding the sensitivities of their energy use data, particularly when deciding whether to share it with third parties. Opportunities exist for accidental or deliberate surveillance via energy use data, and these need to be identified and managed. Prototypes represent a useful tool for navigating this space, and we provide considerations for how they can support users in speculating over possible energy futures.

CCS Concepts: • **Human-centered computing** → **User centered design; Visualization design and evaluation methods; User interface design**; • **Security and privacy** → **Social aspects of security and privacy**;

Additional Key Words and Phrases: Energy, electricity, eco-feedback, prototype, energy data, interpersonal, IoT, smart home

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1 Introduction

The recent growth of smart metering, **Internet of Things (IoT)** technology, and energy monitoring systems underpin ‘energy transitions’ worldwide, heralding a new age of more freely available residential energy-use data [14]. Energy monitoring systems for household energy use and rooftop solar production (hereafter: ‘eco-feedback’) are proliferating, where the global Home Energy Management System market (including **in-home displays (IHDs)** and interface devices) was valued at over USD\$180 million in 2020 [67]. Eco-feedback IHDs are offered free of charge with every smart meter installed in the United Kingdom,¹ which are being rolled out at a rate of over three million installations per year [5]. Techno-optimistic assessments of energy transitions position energy use data as beneficial to network service providers [86], customers [26] and energy businesses [80]. Forthcoming regulations in Australia seek to maximise these potential benefits through more seamless data sharing [41]. Yet the nuances of household energy use data, families’ relationships with this type of data and the possible future trajectories for how it will be (or could be) shared are not fully understood. ‘Surveillance capitalism’—the extraction of behavioural data for onsale to advertisers—continues apace, where frontiers of data capture continue to expand from search history to voice assistants to car dashboards [96]. Although analysis of household energy data can provide insight into appliance usage and otherwise private practices [6, 36], it is not yet comprehensively explored as a source of behavioural data for surveillance capitalism [96].

Eco-feedback systems in households are often unevenly understood by individual household members [11, 87, 94], meaning they share many privacy and consent-related challenges with home IoT technologies (e.g., smart doorbells, IoT cameras and IoT toys). Malicious misuse and exploitation of smart home IoT technologies include corporate surveillance, privacy breaches, spousal/child monitoring [7, 65], and even intimate partner abuse, assisted by uneven knowledge of the technology’s capabilities [61]. Legislative instruments mostly favour interpretations of data as ‘individually owned’, which can be suitable in some cases (e.g., healthcare, fitness data, personal banking) but can be inadequate to cater for the nuances of ‘*interpersonal*’ data [45], i.e., data, where multiple individuals are simultaneously implicated, such as IoT cameras, smart doorbell logs, and increasingly, household energy use data. Open questions remain about ownership and rights pertaining to interpersonal data, particularly in cases where co-habiting arrangements change, such as a family separates or a shared house disbands [40, 45]. Answering these questions is made difficult by the limited public understanding of what inferences are possible through analysis of energy use data and the substantial effect of factors such as data capture frequency and local knowledge on the clarity (and hence sensitivity) of these inferences [12]. The average Australian has limited energy literacy [52] and typically considers their energy use only when the bill arrives [87]. This means that without substantial scaffolding, many users are unlikely to be aware of the potential sensitivities and implications of high granularity energy use data and might struggle to provide informed responses to considerations around control, sharing, and ownership. Accordingly, the aims of this article are to: (1) Understand householders’ concerns and considerations around privacy, consent, data sharing, data ownership and surveillance relating to energy use data and eco-feedback; and (2) Identify necessary human-centred design supports to address these concerns and enable a more consensual smart energy future.

Toward these aims, we developed Household Wattch, a speculative *provotype* [16] that reimagines energy use feedback. Household Wattch profiles households according to their energy use. Through analysis of their energy use data, it presents to households inferences related to their insurance risk, sleep patterns, appliance usage, and occupancy. We presented Household Wattch to 16 Australian families to explore how users conceptualise the sensitivity and visibility of household practices through their energy use data. The speculative design framing is useful for rapidly acquainting

¹ <https://www.edfenergy.com/smart-meters>

participants with the near-future of more ubiquitous, shareable and commercially valuable energy data. Unlike other speculative works related to energy use [75, 82, 93], Household Wattch is based on each participating household's own energy use data (rather than hypothetical or synthetic data). Through extended conversations, Household Wattch elicited considerations about interactions with energy data across three scales: Within the household, Beyond the household (e.g., data interactions with third parties) and Post-household (i.e., when a household re-configures or disbands). The contribution is a series of design considerations around interpersonal energy use data in households, addressing 'open questions' regarding data ownership, legacy and consent [40]. We highlight opportunities for design and policy to protect the interests and privacy of users in the near future of more available, attributable and sharable energy use data.

2 Background

2.1 Eco-Feedback in the Home

Eco-feedback refers to digital feedback on energy use [42]. It is also referred to as 'energy use feedback' and 'IHDs' [23] and has been studied (in its various forms) for decades [81]. Prior research investigates the potential of eco-feedback to engender energy savings among households [31, 38, 68, 81]; to maintain householder interest over time [50, 51, 72, 83]; identify what design attributes and visualisations best maximise engagement and encourage energy reduction [28, 42]; and, how families adopt, domesticate, navigate, and share learnings from eco-feedback [11, 46, 53, 84, 87]. Earlier studies found eco-feedback can lead to average household energy savings of 5%–12% [1, 38], though user engagement decreases over time [51, 83]. Interest in eco-feedback can be limited to a single household member, to the exclusion (or disinterest) of others [23, 84], and initial energy savings are not always sustainable in the long term [23].

Beyond energy savings, eco-feedback is recognised as a tool for education and energy literacy [79] and a conduit to family conversations around energy use with children [34, 46]. Eco-feedback can engage children deeply in household energy use matters, even when not designed to do so [25, 84]. However, there are limited studies about eco-feedback that include perspectives from multiple members of a family unit [11, 51, 84]. Eco-feedback is unevenly understood between household members [11], and often managed as part of existing divisions of household labour [84]. Explorations of storage, sharing, consent, understanding and responsibility for energy data are limited [82].

2.2 Interpersonal Data

Energy use data when measured from a multi-occupant household is a form of '*interpersonal*' data [45], i.e., multiple users' actions contribute to a single data stream [7]. In this respect, it shares similarities with other smart home technology such as IoT cameras, which record actions of multiple users simultaneously [74], or smart assistant query logs, which detail who has asked Google/Alexa which questions [47]. **Human-Computer Interaction (HCI)** studies of IoT data in-the-home explore agency, literacy, evolving social norms around monitoring, and acceptance of passive surveillance [20, 47, 74, 77]. There is a growing recognition of the privacy implications of home IoT devices [65], including surveillance opportunities and the '*datafication*' of household members, which is enabled through their interactions with smart home technology [47, 48, 66]. The distribution of agency in many smart home deployments is unequal among residents, where the 'pilot' (i.e., instigator of the home IoT) typically has greater knowledge, control and access to the system's functionality than other occupants [44]. The same is true for eco-feedback, where the system is typically installed by—and remains the remit of—one (often male) household energy enthusiast [88].

2.3 Energy Use Data: The Next Frontier for Surveillance Capitalism?

Beyond the home, IoT is identified as the next frontier for ‘surveillance capitalism’—a critique of the pervasive technology-mediated capitalism that is now established across world markets [96]. Surveillance capitalism relies on the capture of personal data that is extracted from users’ online interactions and sold to advertisers, defined as ‘behavioural surplus’ [97]. Supply routes for behavioural surplus have quickly expanded from search engine queries to listening personal assistants, navigation applications, doorbells and home thermostats (among others) [97]. The success of this operating model is its (now) ingrained nature in society, where: ‘...the same channels that we rely on for daily logistics, social interaction [and] work [...] now double as supply chain operations for surveillance capitalism’s surplus flows’ [98]. Processes of normalisation have cemented this form of capitalism in society, where users increasingly accept and expect to be profiled and targeted by advertising [98]. Receiving targeted advertisements (for some) constitute a useful—even pleasurable—experience arising from being targeted by advertising, e.g., feeling personally ‘seen’ and understood by the market [89].

Energy use data is distinct from other sources of interpersonal data such as home assistants (e.g., Amazon’s Alexa; Google Home Assistant), IoT cameras or doorbells because the sensitivity of the data depends strongly on the frequency of data capture, which varies widely between energy monitoring devices. At common smart meter² data capture frequencies (15–30 minute granularity) it may be possible to estimate household occupancy patterns (e.g., working from home) and approximate location [29], but no inferences are possible into individuals’ actions. Whereas at the 10–60 second granularity of many commercial energy monitors³ [6], it is possible to identify certain appliances such as fridges, toasters, microwaves and heating/cooling activities [36]. The value of energy use data as a basis for targeted marketing is becoming well utilised. Companies such as Bidgely offer customer load disaggregation services for electricity retailers, stating on their website: ‘Similar to Netflix and Google profiling their customers uniquely one at a time, we enable energy, gas, and water utilities to accomplish the same level of personalized profiling’⁴. Yet many users lack appliance-level energy literacy [21] and may be partly or wholly unaware of the extent of profiling and segmentation possible from analysis of high granularity energy use data [82].

2.4 Design and Legislative Challenges for Interpersonal Data

Flintham et al. [40] highlight two key challenges for design involving interpersonal data in the home: (1) Informed consent from all household members is complicated by the inadequate provision in existing legislation for interpersonal data (where multiple individuals are simultaneously implicated). (2) Open questions remain about control and ownership of interpersonal data in household transitions or how ‘right to be forgotten’ may be executed in situations where it is not actually possible to redact the actions of a single person from an interpersonal data stream [40]. The legitimacy of interpersonal data as evidence in court is (to date) unclear but may be plausible [92]. For example, the Depp-Heard defamation trial set a precedent in the United States for the use of metadata from messages and photographs as evidence in court [64].

Children are often poorly catered for with respect to consent for interpersonal data. In Australia, informed consent for children aged under 18 is largely the responsibility of parents or guardians [65, p. 210]. Which places a considerable responsibility on parents and guardians to determine when their children can/should begin to provide consent themselves and to act in their children’s

²By “Smart meter” we mean smart-enabled household electricity meters located in the meterbox and used for billing. This is distinct from energy monitoring (see footnote 4).

³By “energy monitor” we refer to IoT solutions separate to the household electricity meter that provide data visualisations through eco-feedback displays or IHDs to the household on their energy use and/or solar generation.

⁴www.bidgely.com

best interests with respect to privacy [65]. For example, at what age does a baby monitor transition from a tool for safety and comfort into an unreasonable invasion of a child's privacy? [44]. Older children are afforded a level of privacy where those aged 15 and over can independently open a bank account, obtain a Medicare (Australian healthcare) card and, if they wish to, remove themselves (and hence their future medical history) from their parents' Medicare card [10].

Energy use data differs from many other forms of data in that attribution is dynamic. Namely, the visibility of a specific person's actions in energy data varies greatly depending on granularity of data capture and local knowledge, such as the number of people at home at the time [36]. Yet because energy use is a single data stream, it is not possible to redact an individual's actions without deleting the entire dataset, which is unlike smart assistant queries where each user has control over their own query history. Regulatory changes in Australia's energy market mean that smart metered energy data will soon become more granular and more freely available. Australia's smart meter capture and transmission frequency will gradually increase from 30-minute to 5-minute intervals [2]. Five-minute smart meter data capture is thus far unprecedented in an international context, where (at present) all of Europe, the United States, and elsewhere are restricted to 15–30 minute data capture [2, 90]. The forthcoming **Consumer Data Right (CDR)** legislation in Australia is designed to remove barriers to sharing energy data, namely to: *'...enable consumers to authorise third parties to access their energy data....more easily share their data to get a better deal on a range of energy products and services'* [32]. Yet the average consumer still only interacts with electricity once per month when the bill arrives [87], raising a question of their ability to recognise potential sensitivities in their energy data. These factors make Australian customers an ideal cohort to base this present research into the insight possible through energy data analysis to: (a) ensure users can positively benefit from this likely near future of more sensitive, more freely available and more sharable energy use data and (b) highlight regulatory and policy priorities to ensure agency, awareness and safety are preserved.

2.5 Acquainting Users with Possible Energy Futures: Prototypes, Provotypes and Speculative Design

In a rapidly changing energy landscape of smart metering, demand response, household-energy market participation and energy-monitored data, HCI has played a pivotal role in acquainting and preparing users for different possible energy futures [4, 30, 39, 75, 82]. Extending well beyond eco-feedback, this large body of work comprises design fictions that explore more distant or speculative energy futures and prototypes designed to provide users with experiences of nearer-future energy technologies such as battery scheduling or tariff switching. Prototyping works by Costanza and colleagues [4, 30, 39] seek to understand how *'...autonomous interactive computing systems may mediate user interaction with future energy infrastructure'* [4]. These include an agent-based home battery charging scheduler which aimed to probe household's acceptance of AI in scheduling laundry in the (then) future of cost-reflective electricity pricing [30]. Findings highlighted the tensions of deploying rationalistic systems to schedule domestic practices [30]. A separate study found distrust and unfamiliarity featured as central concerns affecting users' willingness to utilise automated and semi-automated recommender systems for electricity retailer and tariff switching [39]. These high-fidelity functional prototypes are instrumental in eliciting more authentic and informed user perspectives of near-future interactions with electricity [4, 30, 39].

Speculative design is an ideal tool to immerse users in possible future scenarios for which they lack lived experience and to elicit authentic responses about their values and perceptions in such scenarios [35]. Through providing rich contexts to scenarios, speculative design overcomes the difficulties of obtaining accurate responses using hypothetical methods (e.g., Discrete

Choice, Willingness to Pay), which require users to pre-empt their likely actions in a future situation with limited additional context [55]. Few examples of speculative design in the energy data space exist. Those that do typically explore more provocative futures and challenge existing systems of energy provision and delivery. Prost [75] conveyed different potential affordances under different future energy concepts with energy using ‘speculative hairdryers’. Kwon et al. [60] explored personal accountability for resource use in the home through an IoT showerhead prototype which recorded and relayed multiple details to householders of water use and shower length, demonstrating the ‘uncomfortable’ degree of legibility in water use practices when data was interpreted with local knowledge. Neighbourhood Wattch explored users attitudes towards a future where every home’s energy use is visible and publicly available on a neighbourhood scale [82]. It found users do not immediately consider energy use data as sensitive and highlights the substantial effort necessary to better acquaint users with the ways in which household practices can become legible through analysis of energy use data [82]; a central focus for this present study.

Provotypes (provocative prototypes) are a future-focused prototyping method that embodies tensions surrounding an area of interest for collaborative analysis and exploration of design possibilities [16]. The method represents a response to the observation that ‘...discussions with users about prototypes [as opposed to provotypes] often revolve as much around issues within the current practice as they do around what might be desirable aspects of future practice’ [16]. Provotypes share similarities with Technology Probes [54], recognising that ‘working versions’ of technologies need to be deployed to understand and pre-empt future use. Technology probes focus on understanding future technology potentials, while provotypes focus on bridging the worlds of design and human perceptions [16, 17]. Our choice to aim for a provotype relates to the difficulty in engaging users’ with future concerns around energy data when many users are not even aware of the main contributors to their energy use [52], let alone more complex issues of energy data consent, legacy and manipulation. We also wished to create something of a provocation that causes users to think critically about and question possible energy futures.

3 Method

Energy use data was collected and analysed from 16 households taking part in a longitudinal eco-feedback trial, who all had access to a web-based eco-feedback dashboard for the 18 months prior. An alternative and provocative dashboard based on interpretations from each participants’ own energy use data was created and trailed with the 16 households, called Household Wattch. The specific intent of Household Wattch is to explore issues of ownership, control, and permissions for interpersonal energy use data, aiming to address open questions related to these matters in the literature [40, 45].

3.1 Household Wattch Application Design

Household Wattch is a near future desktop/tablet app that profiles users according to computational analyses of their energy use. It closely resembles a ‘provotype’, given its’ specific intent to provoke, to ‘embody a tension that surrounds a field of interest’ [16, p. 395]. A provotype can enable conditions that create greater engagement and more focused, reflective conversations about the content than a more passive or conventional prototype [17]. Household Wattch aims to embody the tension between eco-feedback as a positive tool for saving energy, with its more problematic possibilities, such as facilitating surveillance and commodification. It re-frames energy data from innocuous to potentially sensitive, another data trail to be commodified. The key purposes of designing and deploying Household Wattch were to: (a) rapidly build users’ understanding of the potential

sensitivity of high granularity energy use. (b) Use this greater understanding to capture users' attitudes, concerns and considerations regarding energy use data sharing, ownership and transfer of ownership.

Household Wattch utilises participants' own energy data, presenting a provocative commodification of households in the style of an interactive eco-feedback dashboard. Households are profiled through their previous month's energy data, receiving scores in four categories: *Home Insurance Risk*, *Household Health*, *Sustainability Index*, and *Appliance Health Check* (described below), which challenge the notion of data framed as '*plain and neutral, thereby objective and, inevitably, true*' [33]. Household's scores across these metrics are derived through analysis of each household's monitored energy use data (detailed in Appendix A). Basing the households' scores on their actual monitored energy use data provides an authentic and provocative experience of having aspects of their lives played back to them through the medium of eco-feedback, which all were very familiar with. Yet Household Wattch differs from most eco-feedback systems because it presents a summary of previous months' energy data rather than visualising real-time data. In this respect it sits between interactive eco-feedback and summaries of energy use, such as electricity bills or Business Energy Reports [57]. Household Wattch benchmarks each individual household against the average of other (anonymous) households within the trial, emulating how normative feedback is already utilised on Australian electricity and water bills [9]. The intent was to provide an illustration of the potential privacy sensitivities resulting from analysis of energy use data, how their energy use profiles may look to others within and beyond the house, and from this experience, use the application as a platform for constructively working towards solutions to these problems.

3.1.1 Energy Use Data Analysis and Back-End. Household Wattch uses 10-second frequency energy use data collected with consent (detailed below) from 16 participants' households, who had already had an energy monitoring system installed in their house, as part of an 18-month longitudinal eco-feedback study. Energy data for Household Wattch was collected using an IoTaWatt energy monitor fitted to each household's electricity meter box. The IoTaWatt measures AC current continuously from all electrical circuits in the meter box, logging 10-second averages for each circuit, creating approximately 129,000 data points per household per day (depending on the number of electrical circuits in the house).

Household Wattch draws from household energy use data and related demographics, including age and ownership of different appliances to profile households against four criteria:

- *Household Health*: A measure of overall health of the household comprised of (1) sleep duration, (2) sleep disturbance (less disturbed sleep is more healthy), and (3) eating habits, i.e., cooking on an electric stove versus microwave (more microwave = less healthy).
- *Sustainability Index*: A measure of sustainability. Comprised of (1) average daily electricity consumption for non-solar owners or average daily grid import for solar PV owners, and (2) a metric titled "judicious heating and cooling", based on a correlation of air conditioning use versus outside temperature to determine whether a household runs air conditioning only during extreme hot/cold temperatures (high sustainability score), or runs it during mild temperatures as well (low sustainability score).
- *Appliance Health Check*: Presented as a future addition, where faults in appliances can be identified through algorithmic monitoring of load signatures and may be useful in warranty claims or appliance service call-outs.

Appendix A provides detail of each of the four Household Wattch criteria and the metrics underpinning them, including analysis method, scoring protocol for each and justifications for their inclusion in the app.

The four criteria (above) represent a deliberate mixture of dystopian and utopian energy futures, in alignment with the original argument of speculative design to provide balance in the depictions of futures [35]. *Appliance Health Check* and *Sustainability Index* highlight the utopian, predominantly positive energy futures of remote energy audits, sustainability tips, and remote diagnosis of appliance health issues [63]. While *Home Insurance Risk* and *Household Health* represent a more dystopian product of remote surveillance and discrimination possible through analysis of personal data [97].

The selection of metrics underpinning each of the four criteria aims to demonstrate the potential sensitivity of household energy use data, but in a playful and non-confrontational way. The metrics are based on findings from relevant literature (e.g., sleep disturbance can mediate mental health [56], and forgetfulness can result in leaving the oven on, which creates house fire risk [37]). Yet they are presented in a way that we hoped households would recognise and find humour in their (deliberately) overly rational assumptions and inferences. For example, sleep disturbance is measured by the average amount of activity on the electrical lighting circuit at night (details in Appendix A). This is a plausible, but also very crude approximation of actual sleep quality, given one can be awake extensively in the night with the lights off. We also aimed for the metrics to invite reflection on how attribution can vary depending on local knowledge. For example, “oven left on” might be ambiguous in a large household or highly attributable with knowledge that only one person was home at the time. The *Household Wattoch* interface and interviews (below) worked in conjunction, where the interface was designed to create an understanding of the potential sensitivities of energy data, while the interview questions invited participants to contextualise this understanding to their own circumstances.

3.1.2 Household Wattoch Interface Design. The *Household Wattoch* interface drew inspiration from modern eco-feedback dashboards such as Smappee, Solar Analytics and Phisaver. It was designed over multiple iterations, with the first author creating an initial mock-up using Adobe XD (where both screens and interaction flow were developed). The initial prototype was further refined and developed by the second author using Adobe Illustrator and XD, with the final product featuring a plausible, high-fidelity aesthetic (e.g., logo, minimalist, high contrast dark colour scheme) to reinforce authenticity to eco-feedback. The application comprised of a simple overview page (Figure 1) and more detailed component pages (Figure 2). The home screen (Figure 1) shows a dashboard view containing a floating navigation menu (Figure 1A), individual component cards (Figure 1B), profile details of the resident (Figure 1C) and the date range of the dashboard (Figure 1D). Each component is presented as an individual card (Figure 1E) where a card presents the component category name, a corresponding icon, the score of the household for each metric (between 0 and 100), and the individual household score compared to the neighbourhood average (bar graphs, where the yellow line and dot represents individual household score, and white dot/grey line represents the average score among other households for that particular metric). Clicking any of these cards takes the user to a more detailed overview of that metric. The full interactive prototype is available at the weblink in [Supplementary Materials](#).

Clicking on any of the tabs (both in the navigation and overview cards) brings up a detailed page where specifics of the scores, corresponding visualisations, and details of how the scores were calculated, i.e., time-series of energy use data depicting the households’ sleep patterns (Figure 2). Figure 2 illustrates the detailed screen for the *Household Health* panel, showing the constituents of *Sleep Disturbance*, *Sleep Duration*, *Cooking Habits* and *Routine*.

3.1.3 Speculative Targeted Advertisements. To supplement the application, a series of speculative targeted advertisements were created as design cards (Figure 3). The advertisements leveraged familiarity, using ‘bootleg’ (fake, yet easily recognisable) adaptations of popular Australian and

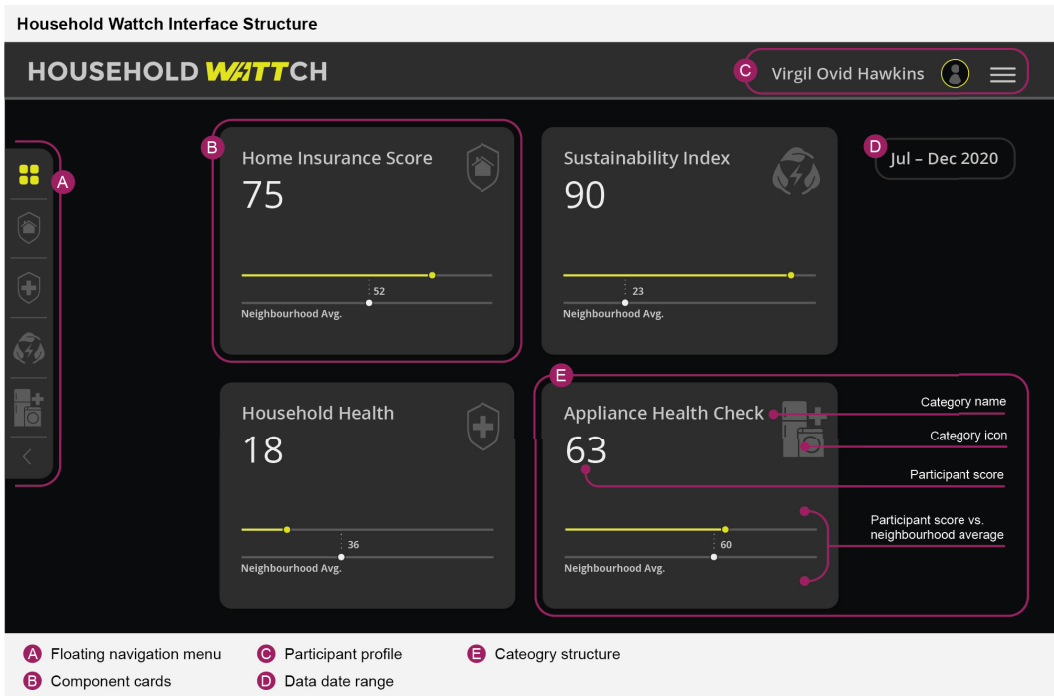


Fig. 1. Home screen of Household Wattch with interface structure; magenta annotations are not part of the interface.

international brands. Two cards were presented to each participant based on inferences made from their scores. For example, households with low sustainability scores might receive the advertisements for smart scheduling plugs (Figure 3A) or solar panels (Figure 3C). This experimentation with materiality of design tools [58] aimed to create a safe space, leveraging humour to allow participants to freely discuss their underlying concerns pertaining to the data and avoid overemphasising invasive/dystopian aspects of Household Wattch. We leveraged *familiarity*, where the semiotics, typography and branding elements in the advertisements were familiar to participants, helping them envision application possibilities through data inferences (Figure 3). This approach to world building experiments by making the familiar strange, i.e., grounding speculative ideas in believable structures and boundary spanning objects [85] that participants can relate to while being flexible enough to allow scrutiny and critique.

3.2 Sampling and Participants

All 16 households who participated in Household Wattch were already participants in an existing longitudinal 18-month eco-feedback trial. The trial originally comprised of 19 households, where 16 had chosen to take part in an optional further interview at the end of the trial. Recruitment for the longitudinal study sought households who were willing to adopt (free of charge) a Phisaver [73] energy monitoring system, including its web-based household energy use dashboard. Participation was advertised through community social media pages. Subscription was constrained to 20 households based on budget for energy monitor purchase and install. One household was excluded owing to an unsafe electrical meter box.

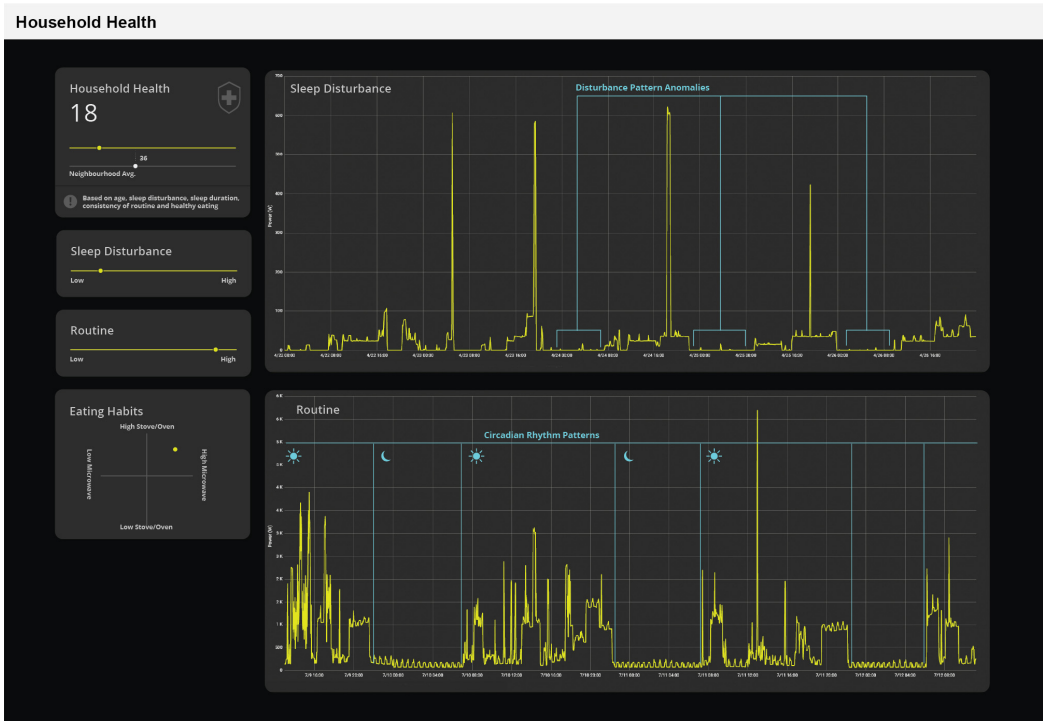


Fig. 2. Household Health screen, detailing constituents of this analysis (note: zoomed in view without top and floating navigation bars, for readability).

None of the original 19 participants were known to the researchers at the commencement of the trial. The sample is, therefore, representative of many longitudinal eco-feedback studies in that there is a likely self-selection bias towards households interested in learning more about their energy use or saving energy [46, 50, 84]. During this 18-month trial participants had been interviewed at three stages (pre-install, 2–3 months after install and 12 months after install), had two short online surveys and had (for a period) received fortnightly SMS messages highlighting their energy use relative to the previous fortnight. Accordingly, participants in the Household Watch interview were very familiar to the researchers, where a level of rapport and trust had been built throughout the original trial. Second, because of their prior participation, the households should be considered more energy literate than the average Australian household and already had a general understanding of the types of inferences possible through energy monitoring, which increases the level of informed consent relative in this process relative to households who had never received an energy monitor.

The informed consent process for the longitudinal trial was undertaken at its commencement. All participants provided written consent to: having an energy monitor fitted to their home, their energy use being monitored, stored and presented on dashboard(s), their energy use data shared with the research team and participating in interviews and surveys, including providing feedback on alternative energy-related prototypes and dashboards (UQ Ethics Number NH03343). Hence, the activities of Household Watch (interviews and energy use presented on an alternative dashboard) were covered under the eco-feedback trial in which the interviewees were already participating. Prior to the Household Watch interview, participants were emailed with an optional invitation



Fig. 3. Full range of bootleg targeted advertisement design cards; two cards were presented to each participant based on their scores for different metrics (e.g., FaultyBlinks mattresses for those with high Sleep Disturbance scores).

to: “take part in a study of an alternative eco-feedback dashboard prototype created from their own energy data”. Interviews were then arranged with the 16 participants who had who chosen to take part in this optional further interview. No incentive was given for the interviews. These households had detailed 10-second circuit-level energy use data available, which they had shared consent (as part of the longitudinal trial). Importantly, this sample also ensures that all participants already had a reasonable understanding of the inferences possible through energy use data, owing to (a) their family co-existing with eco-feedback for the 18 months prior and (b) previous interviews had involved high-level discussions over privacy aspects of energy use data from their energy monitor. Participants were also highly familiar with the researchers, which we felt was important given the nature of Household Wattch profiling users with their own data. Accordingly, Household Wattch was not itself introducing the concept of energy use data attribution or privacy to participants but rather introducing new and more provocative inferences and aggregations that participants may not have thought of themselves.

The household representative contacted was asked to invite other adults to take part (our ethics approval did not extend to children), however only in one household (P6) did both adults contribute throughout the interview. The absence of perspectives from other household members is a limitation of the study. However, we felt that mandating dual participation (rather than simply encouraging it) would inevitably further reduce the already limited sample for whom we had existing relationships with and consent to monitoring and use of their energy data.

Each households’ attributes are detailed below. All were located within the Greater Brisbane area. Household income (before tax) was gathered using the Australian Bureau of Statistics income brackets. Self-reported household annual income ranged from \$50,000–\$75,000 to \$600,000–\$700,000, median \$125,000–\$150,000. This value indicates our participants are broadly representative of the

Table 1. Participant Information Including Age, Employment, Income and Ownership of Specific Appliances

ID	Ages of residents	Employment of adults	Household income (before tax, per annum)	AC	Solar PV	Pool	Dryer
P1	40, 36, 6, 1	F. Research scientist * M. Nurse	\$150,000–\$175,000	✓	✓		✓
P2	38, 36, 6, 6	M. Engineer *	\$125,000–\$150,000	✓	✓		
P3	49, 46	F. Home duties M. IT professional *	Declined to respond				✓
P4	41, 37, 3, 1	F. Home duties M. Public servant *	\$200,000–\$225,000	✓	✓		
P5	44, 43, 5, 7,12,14	F. Teacher M. IT professional *	\$175,000–\$200,000	✓			✓
P6	40, 37, 13, 12, 9	F. School teacher M. Software engineer *	\$100,000–\$125,000	✓			✓
P7	25, 27	F. Part time supply teaching * F. PhD student *	\$50,000–\$75,000	✓			
P8	52, 50, 17, 14	M. PhD student F. Project manager-telecoms *	No data	✓	✓		✓
P9	60's, declined, 24	M. Civil service-procurement F. Part time Lecturer *	\$175,000–\$200,000	✓			✓
P10	56, 49, 17, 15	M. Snr Policy advisor Grandchild. Student M. Retired academic *	\$600,000–\$700,000	✓	✓	✓	✓
P11	47, 39, 4, 4, 4	F. Company director F. Public servant-part time *	\$200,000–\$225,000	✓	✓	✓	✓
P12	50, 46, 14, 10	M. Self-employed F. Self employed *	No data	✓	✓		✓
P13	30, 30	M. Self employed F. FT- Counsellor *	\$100,000–\$125,000	✓			✓
P14	38, 38, 13, 11, 8, 6	M. FT- Masters Student M. Researcher *	\$75,000–\$100,000	✓		✓	✓
P15	55, 46, 19, 15	F. Stay at home mum M. Database admin *	\$150,000–\$175,000	✓			✓
P16	33, 30, 1	F. HR payroll Adult child. Temp in HR M. Procurement officer *	\$75,000–\$100,000	✓			
		F. Maternity leave (Town planner)					

*Household representative who participated in the study. Note that both adults in Household 6 participated in the interview.

Australian mean weekly gross household income (2020–2021) of \$2,329 (\$121,108 per annum) [8]. Table 1 lists details of each participant and appliance ownership. It should be noted that all households were variations on nuclear households, e.g., domestic partners living together with or without children or grandchildren.

3.3 Interview Protocol

Household Wattch was evaluated through an extensive semi-structured interview allowing participants to explore the interface themselves. The static “summary” nature of the dashboard meant there was little benefit to “deploying” it with households, as the interface does not update regularly.

Semi-structured interviews were conducted with participants online over Zoom. The interviews began with a brief introduction to Household Wattch, which was described as an alternative energy use feedback dashboard (relative to their current eco-feedback dashboard [73]). We clarified that the purpose of creating Household Wattch was to communicate the potential for algorithmic profiling through energy use and discuss energy use data in greater depth. The research team then facilitated a brief click-through of the interface, describing the main features. Participants were then given control of the cursor to navigate and explore the interface themselves and were encouraged to ask any questions or seek clarifications about the score, data and inferences. This approach enabled the participants to lead the discussion in the initial exploration/familiarisation period, followed by the research team asking questions related to:

- Initial thoughts and reactions to each dashboard constituent, e.g., *“How do you feel about being profiled in this way?” “How accurate are these inferences of your household?”*
- Permissions and legacy, e.g., *“Which householder should respond to permissions requests for this type of data?” “What should happen to this data if you were to break up or move house?”*
- Privacy/utility trade-offs, e.g., *“Would you share this data with a health insurance company if you were offered a lower premium for high health scores?” “Would you share all your data?” “Is there any data you would want omitted?” “Is anything in this dashboard too intrusive for you to share?”*
- Questions on the targeted advertisements, e.g., *“What are your thoughts on receiving targeted advertising that you suspect might be based on your energy data?” “Do you have any positive/negative feelings?” “Would you still cash in these vouchers even if they were based on incorrect inferences?”*

The targeted advertisement design cards were presented to the participants after most of the interface-specific discussion had taken place and broader privacy implications were being explored. Each interview was facilitated by two researchers (first and second authors), where the first author who had a strong rapport with the participants lead the facilitation, and the second author documented discussion notes. On average the interview duration was 45 minutes, and all interviews were recorded, and transcribed. The full question list is available in [Supplementary Materials](#).

3.4 Analysis

This article’s aim is to address specific gaps in the literature (described in Section 1), and hence interview questions were designed to elicit responses to these gaps. Analysis was comprised of a two-step process involving (1) collating and categorising participants’ responses to specific questions and reporting these directly before (2) undertaking a thematic analysis adapted from Braun and Clarke [19] to identify emergent themes not captured through the interview questions. Both are described below.

Response categorisation: First, we collated and categorised participants’ responses from interview transcripts and notes, including (1) participants’ general feelings towards Household Wattch and perceptions of its accuracy, (2) which household member should be responsible for granting permissions to share household data, given energy data simultaneously implicates all householders, (3) how data ownership might be managed in the case of household dissolution (e.g., family breakdown, divorce, or occupants leaving the household). Responses were collated, tabulated and

compared by the first and second authors. Findings were then tabulated according to the scale(s) at which they were described by participants: Personal scale, Household scale, Beyond the Household and Post-household (descriptions below).

Thematic analysis: Second, we conducted a thematic analysis process, adapted from Braun and Clarke’s “most recent articulation” of their thematic analysis process [19]. This process involves a six-stage process for engagement with data, coding and theme development, comprised of: “1) *Data familiarisation and writing familiarisation notes*; 2) *systematic data coding*; 3) *generating initial themes*; 4) *developing and reviewing themes*; 5) *refining, defining and naming themes*; and 6) *writing the report*.” [19, p. 331]. It is acknowledged that the approach is not intended to be followed rigidly and that stages may blend together as analysis becomes more recursive with time [19].

After responses to specific questions had been tabulated (above), the lead author studied transcripts to identify reactions, positions, revaluations, refutations, or explanations from each interview that extended beyond responses to the questions posed. These responses were printed onto cards, generating 237 unique responses, averaging 16.9 responses per participant. Once compiled into cards and devoid of the context in which the responses were made, the cards were placed on a large boardroom table, and the six steps of Braun and Clarke [19] were enacted, with three researchers (co-authors) participating. (1) Each author independently engaged in individual data familiarisation and writing notes, followed by (2) systematic ‘semantic’ coding [24], i.e., the authors drawing upon explicit statements from participant responses—to make sense of the data. The second and third author had no additional knowledge about participants, hence their analysis was based purely on critically evaluating the data from the transcripts alone. The first author, however, had a longstanding relation with the participants, with multiple prior participant interactions as part of the longitudinal eco-feedback study the author was leading. This resulted in additional contextual knowledge about the participants. They were, hence, able to draw additional inferences about the statements in the data transcripts as part of their initial coding.

After the initial individual coding process, all three researchers had a collaborative analysis workshop to discuss and collate the generated codes into thematic clusters collectively—i.e., (3) generating initial themes from the coded and collected data. Here, the first author was able to provide additional contextual knowledge about participants (where relevant), to append the initial individual analysis—adding more nuance, imbuing aspects more in line with ‘latent’ coding [24], i.e., facilitating the discussion so that the collaborative analysis could go beyond the descriptive level. The authors felt this transition from individual to collaborative and reflexive analysis was necessary to explore multiple assumptions and generate richer interpretations of meaning in the data, adding to the rigour of analysis while also acknowledging and managing potential biases held by the first author due to their closer relationship with participants. This was followed by iteratively (4) developing and reviewing themes, where an initial 15 themes emergent through this process were later merged into four key themes and related sub-themes.

Steps 5 and 6 involved further refining themes, leading to a total of seven themes and 11 sub-themes, which represent sub-headings of the results (below). These themes were then assigned to one of three scales in which they occurred. All scales are relative to the “household”, a central unit of analysis for studies of energy use, eco-feedback and energy use data. These three scales comprise:

- *Within the household:* Referring to interactions between eco-feedback and householders. A vast body of literature discusses energy use decisions within the home, how eco-feedback affects decisions in the home and the possible tensions between individual motivations for energy savings and household dynamics [46, 49, 84].

- *Beyond the household scale*: Referring to the sharing or transacting of household energy use data with actors beyond the household. Sharing or selling household energy use data with energy markets, energy businesses or other commercial entities opens new opportunities for households, yet simultaneously raises implications for data privacy [27].
- *Post-household*: Referring to the legacy considerations for household energy data following a household reconfiguring or disbanding. While provisions for data ownership and transfer exist for individual transitions, e.g., childhood to adulthood, or managing someone's data following death [78], provisions for interpersonal data following household transitions are largely unexplored [40].

4 Results

Interactions with Household Wattch typically followed a path of: (1) Initial exploration and curiosity, (2) sensemaking, which involved reconciling the data against local knowledge and beginning to see their families' actions within the data and (3) beginning to understand what their household's data might look to others—including advertisers—without this local knowledge to contextualise it. These realisations then informed discussions with researchers around sharing energy use data, permissions, ownership and legacy considerations of the data. In this section, we present results according to the scales described above: Within the household, Beyond the household and Post-household, noting the responses are (except for P6) perspectives of one adult household representative.

4.1 Within the Household

Sentiment and sense-making: Despite the potentially sensitive inferences it drew from participants' own energy data, all but one (P3) of the 16 households responded positively to the idea of Household Wattch. The provotype was widely conceptualised as an extension of the eco-feedback they were familiar with, but one which generated new and interesting insights about their household from their energy data. Perhaps accordingly, the provotype was oriented as assistive in nature rather than harmful: *"I think it's trying to do something positive, like kind of reduce energy consumption, [it] will get people to be more mindful of how they are using energy, that kind of thing"* (P1). *"I'm loving what I'm seeing so far - this is amazing. (laughs)"* (P5). P14 summed up the sentiment of many, with initial surprise at the level of inference followed by an eagerness to learn more. *"Oh my gosh they know everything about me!! No, the overall reaction is I'm pretty impressed with what you've got there"* (P14). P3 was the only participant entirely uninterested in Household Wattch. P3 was unconcerned about the privacy implications of their energy data but felt the scores were too abstracted from the data itself and would have preferred the data in the rawest possible format to enable his own analysis.

Reflections and realisations on family life: The basis of Household Wattch in participants' own energy use data created legitimacy. Confidence in the authenticity of the data itself meant participants recognised instances where the (purposefully crude) computational inferences drawn from the data departed from their own understanding of their household. For example, P6 and P13 did not operate a microwave and speculated how the "microwave" use reported in the Stove vs Microwave analysis of *Household Health* could be their coffee machine (P6) or blender (P13), reasoning these appliances would likely draw similar amounts of power to a microwave and were often used around mealtimes. P4, P6, P7 and P11 all derived humour from Household Wattch classifying their household as having little sleep disturbance, when in reality all reported their sleep was often disturbed, but in ways that did not involve turning on lights or lamps. These included problematic night-time smartphone use (P7), disruption by children (P4, P11) or difficulty sleeping (P6). Pointing to the dashboard, P6 exclaimed: *"We have no sleep disturbance [laughs]! That's not because we don't wake up at night, just*

that we don't use the light!" (P6). P1, P5 and P7 described how their household's very short "Sleep Duration" was in fact a combination of their early rising and their partner retiring late. P8 reflected on how their household used the microwave a lot collectively, but how different family members used it for healthy and unhealthy purposes: "My daughter has gotten into these brownie-in-a-cup things.... Only 30 seconds in the microwave. You'd think we were reheating a cup of tea, but no, we're actually making brownie in a cup!" (P8).

Realisations made through Household Wattch were often related to the actions of other family members, exemplifying the legibility of practices with local knowledge and the considerable surveillance potential of energy use data. P5 was pleasantly surprised at their *Sleep Duration* metric on Household Wattch, deducing that his teenage son must not be staying up as late as he suspected. P8 came to the opposite conclusion, laughing: *"it looks like you've just busted my 16-year-old for being out of bed!" (P8). P7 mentioned that she would be interested in Household Health to see her partner's approximate sleep time, noting she was often unaware of how late he stayed up studying. P2 likened the utility of Household Wattch to the IoT wearables they already used for sleep and movement tracking for their pre-school-aged children, which paired to the parents' smartphones. P10 recalled an incident during the original eco-feedback trial where he had logged in from abroad during an extended vacation to check the pool pump's operation but realised he was also observing his mother-in-law's practices while she was house-sitting: "occasionally [name] was coming in, I could see when they're using the air conditioners, and that sort of stuff. I didn't do it [log in] for that reason, but it's how I caught that the pool pump wasn't working..." (P10). P5 remarked of the Household Health metric; "This one for me would seem immediately actionable for us. [...] You can see over time how we are tracking, if it's school holidays or if it's during a busy semester if our sleep duration changes over time. And I think the eating habits could be some home truths as well" (P5).*

These findings highlight how Household Wattch was conceptualised as a useful extension to traditional eco-feedback, which provided new opportunities for reflection about family habits and energy use. Equally however, the findings also identify the significant degree of legibility of family members' practices and opportunities for casual surveillance that is possible with local context.

4.2 Beyond the Household Scale: Household-Third Party Interactions

Additional to identifying practices within the home through their energy data, Household Wattch caused participants to consider what their household's energy profile might look like to others or third parties beyond the home who lacked their own local knowledge. The *Home Insurance Risk* metric, for example, surprised several participants who realised their own actions were affecting their insurance risk (according to Household Wattch). Insurance risk became more personal than the usual external determinants such as neighbourhood characteristics, flood zones or crime rates: *"I've just gone through the process of trying to get home insurance because apparently I live in a high flood area and they wanted to increase my premiums by 66%....But you're thinking [less] break-ins because [the electricity use data shows] there's more people working at home here during the day... That's interesting" (P8). P10 realised their "Oven left on" checkbox was highlighted and that this was contributing to their unexpectedly poor Home Insurance Risk score: "Oh sh*t! Y'know, there were about two or three instances where we left the oven on... I did it!" (P10).*

Sharing and permissions for sharing energy data beyond the home: The deeper understandings facilitated by Household Wattch of the analysis potential for high-frequency energy use data factored into participants' considerations around sharing energy data beyond the home and how permissions for sharing this type of data should be negotiated. When asked about the acceptability of sharing their energy use data with third parties to obtain advice or receive offers or discounts (e.g., on appliances or insurance products), three participants gave an unconditional "no" to any form of sharing (P3, P6, P7), three were entirely unconcerned about sharing energy data with companies

and were eager to save money through sharing their energy data (P1, P2, P13). The remaining 11 participants said their answer to any request or opportunity to share would be contingent upon access to the finer details of any such transaction. Of those completely unwilling to share, P7 spoke of the time and management overheads involved in keeping track of loyalty cards, free trials and discount vouchers, while P3 was certain his energy use data would work against him in any sharing transactions, given Household Wattch had identified his irregular sleeping habits. P4, P6 and P8 were not against sharing energy use data, but all mentioned they would have to be satisfied the data they did share accurately represented their household. P6 referred to his earlier suspicions that Household Wattch was misclassifying their coffee machine as microwave use and explained: *"If the insurance company says 'people who put stuff in the microwave are at a greater risk of getting cancer' or something, at the moment, it [Household Wattch] shows that apparently we use our microwave lots. If they base their premiums on that risk factor, then its like 'no, no, no we don't use our microwave lots, we just drink lots of coffee!'"* (P6). P8 was reticent to share data, feeling the Household Health score did not represent their family: *"It might be that one person cooks really healthy food, while the other nukes [microwaves] Uber Eats, you couldn't tell"* (P8). Four participants (P3, P6, P14, P15) spoke of how incentives or discounts could lead users to game the system, doctor their data or avoid being tracked: *"...you know, like shaking your Fitbit to get your steps up so you look better... You can make them say whatever you want them to say"* (P6). P14 considered moving his energy monitor to a different house to improve his data: *"There's people who have those car trackers installed, for car insurance in the UK. They stick the tracker in their Granny's car instead and they drive their sports car around..... I could move the [energy monitor] to a different household"* (P14).

Permissions: Participants were specifically questioned on who should field permissions requests to share household energy use data with third parties. Seven of the 16 participants identified the bill holder as the suitable candidate to field permissions requests to share data, three felt that express consent should be gathered from all adults in the home, and the remaining nine were unsure or unable to answer. P1's response was representative of several participants, highlighting a difference between what they felt *would* happen compared to what they felt *should* happen: *"Legally, I guess whoever pays the bills in a household, because that's a way of getting permission. But personally I feel like it should be up to all the adults in the house"* (P1). P5 stated: *"At the moment, it would be, really, my wife and myself [who should be responsible for consent], because we're both adults, all the children are under 18. But if you have a household with more adults, that is private information about their life... it's starting to get a bit grey there for me"* (P5).

P8 identified the bill holder as the suitable entity to negotiate permissions to share data but explained her children (aged 16 and 18) transitioning to adulthood meant such family decisions were increasingly being discussed as a collective rather than only between the parents: *"Yeah we'd probably have a discussion around the dining room table. I know we talk about the energy monitoring thing [previous eco-feedback trial]. I've shown the kids - the young adults- I've shown them the data that's being generated and how to read it and that stuff and they've shown an interest at the time. So, we'd have a conversation"* (P8). Yet others felt it imperative that consent should be obtained from all adults in a house and not only a single individual such as the bill holder. *"All members of the household should [give consent], because one thing that, probably my biggest takeaway from this, is how personally identifiable your power signatures are."* (P3). *"I definitely think that everyone who has a vested interest in that data should have a say [in how it is shared]"* (P14). P15 was adamant the agreement should legally extend to those over 16, not only adults, pointing to his daughter's choice to redact herself from their family's Medicare (healthcare) card when she reached the age at which this is legally allowed: *"...so I have no idea of whether [daughter's name] has been to a doctor or what she's been to a doctor for"* (P15). P15 additionally pointed to the need to re-negotiate consent clauses as children become of age: *"So at one level you have to give away certain data to the energy company,*

that there are children in the house... it means that when they reach a certain age, the energy company would automatically send out a request to update the data agreement because the kid now has a right to say in their data" (P15). P10 and P15 also pointed to the high level of energy literacy required for informed consent in the first place, which might be difficult to achieve in busy families.

These findings illustrate the complexity of factors requiring consideration when sharing interpersonal energy use data beyond the home. Considerations include how the data may be used and interpreted by third parties, as well as considerations of gaining consent for sharing from others in the home who are implicated in the data. Informed consent may be difficult to obtain from family members who lack knowledge or interest in energy matters, where managing energy is often the domain of only one householder. P10 concluded: *"But ultimately, I think the main thing is to make people aware, because I'm sure there's lots of people who just wouldn't be aware that there are issues there"* (P10).

Corporate surveillance and targeting: Each participant received two targeted advertisement cards (Figure 3) based on their Household Wattch scoring, representing realistic provocations of the types of benefits possible through sharing energy data with businesses. None of the 16 participants rejected the idea of receiving targeted advertisements. Responses ranged from apathy, resignation or indifference (P1, P3, P4, P8, P9, P10, P11, P12), to active interest in receiving ads (P5, P6, P13 and P14). Most considered targeted advertising as a vaguely problematic but unavoidable norm of modern living: e.g., *"we are already bombarded by ads"* (P8), *"it's just part of life"* (P11), *"I'm more concerned about the phone listening to me talking through the mic than my anonymised [energy] data being shared"* (P7). Hence, while not everyone was actively eager to receive targeted advertisements based on their energy data, this form of targeting was not considered any more problematic than their existing situation. P1 joked about the lack of control she felt over her advertisements: *"they can be based on your browsing history, our Facebook things, sometimes even when you're just thinking about things!"* (P1).

The four participants eager to receive targeted advertising based on their energy use data sought tailored information and described the transactional nature of sharing, e.g., wanting something in return for their data: *"I actually don't mind it because you get a lot of good information, sometimes. Because I mean I have been thinking about, you know 'should I be getting a smaller heat pump hot water?' And I just don't really know where to start to even look into that sort of stuff so I guess having those ads pop up [based on energy use data] can kind of help in a way to figure it out"* (P13). P5 was interested in advertisements provided they contained specials or offers, rather than simply advertising, describing his privacy as a strictly monetary negotiation: *[Pointing to one of his two Household Wattch ads] "Where's my discount? What am I getting for [sharing] it? It is not reciprocal... I swipe my [name of supermarket loyalty program] card and sure they know exactly what's going into my basket every week. But I get something back from that. I get some discounts"* (P5).

Together, these findings highlight the degree to which corporate surveillance was normalised by participants. Most fully expected to be profiled by corporations through their data, quite irrespective of their personal stance on the ethics of such practices. The findings show many participants are not opposed to sharing energy data with third parties from a privacy perspective provided there is suitable incentive, transparency, and assurances in place.

4.3 Post-Household—Splitting Ownership, Moving Out, Leaving Your Data Behind You

Addressing an open question posted by Flintham et al. [40], we wanted to understand participants' thoughts on legacy considerations for energy use data *"post-household"* i.e., what should happen to a household's energy use data in the case of a household reconfiguring, moving separating?

"Splitting" assets after a separation: All participants were in domestic relationships and living with their spouse/partner at the time of the interview. None of the 16 participants had considered

what they would do with any of their shared data in the eventuality of a separation or household dissolution: *"We'll fight it out in the settlement!! (laughs)... Um yeah. That is a hard one.... I guess you'd have to say it belongs to both and if you can't agree then it goes to no one..."* [Interviewer asks]: *Do you mean it just gets deleted?* [Reply]: *"I guess so yeah. I don't know how else you'd do it..."* (P6). P8 remarked: *"That's a tough question. That's kind of like the shared Google account who gets the Google account who gets the Netflix account? [...] I don't know the answer. I suppose whoever took the most amount of interest in it..."* (P8).

Others felt that, in reality, the importance of energy data in such a situation would be eclipsed by other considerations, and they would simply not concern themselves with such matters: *"How do you separate energy usage?... You're splitting assets, finances, if you've got kids you've got the awful situation of working that out... No it [energy data] would be right down the bottom of the list"* (P5).

The low level of privacy/sensitivity attributed to household energy use data contributed to participants' feeling they would be apathetic in dealing with such data in a divorce or change in household composition: *"...it's not my financial data it's not my personal relationships or sexual data or anything like that] it's just energy consumption, I don't hold a personal or emotional attachment to it"* (P12). *"I dunno. The care factor, for that would probably be low in an event of that [divorce] happening. I haven't really thought about that, to be honest and probably goes for a lot of other types of data"* (P1).

Post-household surveillance: Opportunities for both accidental and anti-social surveillance were raised by different participants, such as if a family member leaves the household but retains access to energy use data or if a departing home-owner forgets to remove their energy monitor. P8 mentioned how they might easily forget to turn off the energy monitor if they were to move or rent the house out, meaning the incoming tenants would not necessarily be aware of—or consent to—their energy data being monitored. P3 rationalised that he would not like a previous occupant's (detailed) energy use data *"...because that means the person moving into here [after me] must be getting my data."* (P3).

The likelihood of energy monitoring being forgotten about in a breakup or divorce or moving caused P14 to speculate on potential avenues for non-consensual surveillance, such as him retaining access to the energy use data after he had moved out: *"Say if, God forbid, me my wife and I get separated and I become a horrible abusive person and a stalker.... I haven't thought about exactly what the harm from that historic data could be, but the ongoing data [monitoring] could be, you know, quite sensitive"* (P14).

P8 raised the "grey area" of spousal abuse and whether energy data could be used as evidence, such as what activities took place in the home on the day of a specific incident. *"If that was the case, you might not want your data shared at all"* (P8). P10 also identified this type of scenario, but wasn't sure how it could be easily avoided: *"It's probably not on people's priority list to deal with those sorts of issues. And it might be the opposite. Some people might see it as an opportunity to get back at someone. The problem with that is the practicalities of how you stop that stuff? And you can't send a message every month, saying 'is this still your situation?' I dunno how you deal with that."* (P10).

Cumulatively, these findings highlight the substantial surveillance potential of energy use data and how it could be weaponised in a family breakdown. Organising transitions for data control or ownership is understandably unlikely to be prioritised—if considered at all—during such major life events.

5 Discussion

The rapidly developing landscape of energy monitoring, energy data sharing opportunities [41], consent legislation [40] and the expanding reach of surveillance capitalism [96] point to the importance of supporting users in navigating this emerging energy future. The substantial surveillance opportunities through high-frequency energy use data point to the ever-increasing diversity of

digital traces left behind in everyday life [77]. Sitting in between traditional eco-feedback dashboards [26, 42] and more exploratory speculative designs for energy [75, 76], Household Wattch catalysed rich discussions with participants around energy use data in the context of surveillance and commodification, providing an alternative framing for eco-feedback.

The contributions of Household Wattch are three-fold. (1) Technically, it represents a proof of concept for detecting practices, activities and other intimate information (e.g., cooking, sleeping, health) from partially disaggregated 10-second energy use data. It builds on existing computational work exploring the limits of inference possible through electricity data analysis such as **non-intrusive load monitoring (NILM)** [12, 36, 71] through its focus on detecting human practices (e.g., sleeping, cooking) rather than simply detecting discrete appliances [6]. It also demonstrates the depth of insight possible through very basic computational analysis without using NILM or machine learning techniques, which are commonly applied to energy use data [12]. (2) Methodologically, Household Wattch extends speculative design and provotyping for smart energy futures [75, 82, 93] by incorporating participants' own energy data in the interface, culminating in a provocative yet highly personalised and legitimate experience in which householders saw their energy use data through the eyes of third parties who may benefit from it. Household Wattch blends speculative futures with present day realities, where third-party analysis of energy use data and increasing data capture rates are already in place in many electricity networks [29]. (3) Empirically, Household Wattch bridges the respective literatures on within-home surveillance opportunities of smart home devices, such as smart doorbells and cameras [7, 44, 95] and the corporate surveillance implications of sharing data with third parties beyond the home [3, 69, 96]. It identifies the potential alternative roles of eco-feedback beyond energy savings and demonstrates the challenges of designing for legacy considerations of interpersonal data, e.g., navigating control and ownership of energy use data if a household reconfigures or disbands and the imperative to avoid anti-social surveillance. We discuss these factors in terms of their implications for designs involving energy use data (Section 5.1) before reflecting on the design of the provotype which catalysed these findings in the first place (Section 5.2).

5.1 Implications for Designs Involving Interpersonal Data

The findings point to several considerations necessary when designing—or using—technologies that involve the capture of interpersonal data. The four implications below are areas we feel deserve immediate attention:

Implication 1: Eco-feedback should support a range of applications beyond energy saving: Despite a wide range of methods, designs and applications, much of the legacy of eco-feedback deployment and research is framed by the end goal of motivating energy savings through better awareness [1, 25, 31, 42, 46, 50, 81]. By backgrounding energy saving as an explicit goal, Household Wattch demonstrates how energy use data and eco-feedback can be leveraged to foster broader reflection on the production of their family life, on health, sleeping patterns and situated awareness [22]. It caused new reflections on their own and their families' lifestyles (e.g., eating choices (P6, P11, P12), sleeping habits (P4, P5, P7) or sustainability (P9, P11)), yet also allowed insight into what their data might look like to third parties with potential interests in it. Because users can value and learn from alternative representations of their energy use data, designers should look beyond energy savings as an end goal for eco-feedback deployments and experiment with how alternative representations of energy use data can support reflection on family life and might support behavioural nudges toward better sleeping habits, eating habits, or fostering greater understanding of privacy boundaries within a family.

Implication 2: Assist users in navigating the complexity of sharing interpersonal data: Sharing interpersonal data beyond the home is complex, involving negotiation of sharing terms with the

third party, as well as negotiating consent for sharing from those within the home who may be co-implicated in a dataset. The findings support literature suggesting privacy is fluid, negotiated and conditional [62, 69]. Though some were unconcerned about exchanging data or privacy for monetary benefits (e.g., P1, P2, P13), for most, willingness to share energy data was entirely dependent on context, i.e., what data, with whom and the perceived risks and rewards.

The difficulty with interpersonal data is that consent from other household members in sharing transactions is more complicated than for personal data. The Australian CDR will soon be applied to smart metered energy data, providing for a centralised platform to manage consent requests for sharing retailer-metered energy data.⁵ The CDR utilises a gate-keeper single-point-of-contact model of “data owners”. Yet like others, our findings suggest these forms of gatekeeper consent mechanisms may struggle to account for the nuances of interpersonal data [40]. Many participants felt consent from all household members was vital but did not have an easy answer for how this should be achieved. Involving all householders in consent decisions is difficult [91]. Even securing both adults for an interview was problematic, where we invited all adult household members to participate in the Household Wattch interviews, yet this occurred in only one household (P6).

Accordingly, design work is necessary to assist users in better understanding and navigating the sharing of interpersonal data. Challenges include how to manage consent with children and when/how to include them when they become of age. E.g., P15’s daughter had chosen to redact herself (and her health data) from their family Medicare card, as is permissible at 16 years of age. Yet a redaction model is difficult to reproduce with energy data, given that attributability of individuals in energy data is highly variable, depending on data capture frequency [36] and local knowledge (e.g., P5 knowing his son would be the last to sleep at night or P10 knowing his mother-in-law is the only one at home). Hence, it may be impossible to cleanly redact an individual’s contribution to energy data without deleting the dataset entirely. Like others, we observed unequal knowledge of energy use data between eco-feedback “pilots” (lead users) versus “passengers” (e.g., spouses, children) [59] and we expect these nuances of attribution in energy data are unlikely to be fully understood by many passengers in the first place.

Improving consent processes for interpersonal data might involve, firstly, prompting smart home pilots to consider their passengers’ wishes and motivations for sharing/not sharing data. Secondly, better informing smart home passengers about the nature, sensitivities and attributability of energy use data in a way that is not burdensome, given it is unrealistic to assume eco-feedback can overcome the well-documented disinterest from certain family members or transcend established divisions of household labour [87].

Implication 3: Improving awareness of corporate surveillance via energy use data: Being tracked and profiled by corporations is increasingly normalised in society, yet users are seldom aware of the nature or extent [98]. In our interviews, discussions around sharing energy data were accompanied by an implicit expectation that some level of data leakage and corporate surveillance would occur anyway. Energy data was generally considered less sensitive than other data streams such as search engine history or loyalty cards. Several participants were in fact eager to receive targeted advertising, feeling such information would assist them in appliance purchase. Yet awareness of energy use data as a form of “behavioural surplus” [97] is likely to be lower among the general population who has not participated in Household Wattch.

Given the proliferation of devices with energy monitoring capabilities (e.g., smart home IoT, electric vehicle chargers, rooftop solar panels and battery storage) and the emergence of energy use data as a form of behavioural surplus [13], we argue an urgent need to better inform users

⁵Smart metered energy data collected by the retailer for billing is typically 5 or 30-minute granularity and hence much less inferable than the 10 second data used in Household Wattch).

of the corporate surveillance potential of energy use data and understand what inferences are possible at what rates of data capture. This includes (1) gathering a better understanding of the accumulation practices of companies gathering energy use data and what inferences may be gleaned from different combinations of energy use and IoT data; (2) consideration of how best to increase users' awareness of these practices. Household Wattch represents a small step towards designing ways of communicating nuances to users to enable more informed decisions with respect to privacy and consent considerations of energy data. Much future work is required to better communicate the potential third-party uses of energy data and how contextual factors [60, 82] and triangulation with other forms of personal data may affect its sensitivity [43].

Implication 4: Avoiding anti-social surveillance: Technology is implicated in almost 50% of intimate partner violence [61] and often involves monitoring victims via smart technology [18]. While the study was not equipped to uncover any anti-social surveillance among householders in our sample, our findings illustrate how energy use data could feasibly be weaponised. Even in the seemingly innocent examples of “checking up” on children or spouses (e.g., P8 joking Household Wattch had “busted” her teenage daughter for staying up late), it is unlikely the subjects themselves were aware their activities were being tracked. Like others, we observed a large difference in knowledge of the energy monitoring technology between family members [7, 44], and participants freely admitted they would be unlikely to consider data ownership or energy monitoring when moving house or during a divorce and would be likely to be distracted by other priorities.

Technology designers and researchers alike must better recognise and pre-emptively reduce the risk of anti-social surveillance, ensuring protections are in place before surveillance is weaponised. Part of this puzzle is making the surveillance potential known to users in the first place. Literature documents the artful ways teenagers avoid detection by IoT doorbells and cameras [7]. Yet these evasive techniques required the teenagers' awareness of surveillance in the first place. Energy use data represents a far less visible medium of data capture compared to other smart home IoT, such as doorbells and cameras, given the attribution of individual actions varies according to multiple contextual factors. Protections might involve: (a) authority for any household member to request energy data to be deleted, overriding gatekeeper consent for other arrangements, (b) agreement by all household members to a given frequency of data capture upon install of energy monitoring, (c) physical “stop” buttons in the home to allow an individual to pause or stop energy monitoring. Future work should explore and identify the “mosaic” potential for energy use to be co-analysed with data from other household IoT and better unpack the implications for policy. For example, should energy data be permissible as evidence in court, if it was captured during an alleged incident? Can (or should) it satisfactorily capture proof of occupancy or proof of doing an activity such as cooking? How can/should individual attribution actually be proven in energy data? Amazon has recently objected to a warrant issued to access Alexa recordings where it was considered possible that the Alexa might hold evidence relevant to a US homicide case [70], and questions remain as to the ethics and legality of home IoT as evidence in court.

5.2 Considerations for the Design of Provotypes Around Household Energy

Household Wattch enabled rapid sense-making, acquainting users with the multiple inferences drawable from a given type of electricity data. As per the design intent of provotypes, the elicited responses extended well beyond the basic usability of the application interface, forming a bridge between activities of investigation and the design of new possibilities [16]. Being generally satisfied with its function as a provotype, we reflect on three interlinking provotype design considerations based on our experiences:

Legitimacy: Incorporating users' own energy data was central to the legitimisation—and provocation—of Household Wattch. Participants readily identified their families' actions within the data

and many were able to trace how their actions affected certain scores. For example, realising how working from home (P8) or leaving the oven on (P10) affected Home Insurance scores. Hence, we argue that grounding provotypes in participants' actual data (where possible) offers a far more impactful, personalised and engaging experience than using dummy data or simply explaining concepts [60].

Aesthetic and fidelity: Relatedly, achieving legitimization and provocation requires careful consideration of fidelity. We opted for a high-fidelity aesthetic of the dashboard and design cards (e.g., consistent colour scheme, branding elements, Household Wattch logo) to maximise plausibility and believability [15] and mimic the polished aesthetic of the commercially available eco-feedback dashboard participants were already familiar with. Yet despite the high-fidelity finish, we were careful to limit the fidelity of the insights captured—namely, to ensure inferences were plausible but not *too* personal. In adapting the method to different contexts or in designing similarly provocative provotypes, it is critical that consideration is given to this delicate balance of provocation, humour, yet avoiding confrontation.

Provocation ethics: The provocative intent of Boer and Donovan's original conception of provotypes is relatively mild, i.e., to "*expose hidden norms as a basis for debate to rethink sustainability*" [16]. Our motivation aligned with a more active intent to provoke (i.e., to stir up, to cause a reaction⁶); to create an unambiguous statement about the inferences possible through high-frequency energy use, but which does not itself invade privacy. We argue that crafting suitable and ethical provocations require careful consideration of the audience. Household Wattch was tailored to our particular user group, namely, those who had consented to sharing energy use data, who possessed an already high level of energy literacy (owing to the previous eco-feedback trial) and with whom we had already developed a high level of trust and rapport. Despite this, these types of provocative designs, when deployed as part of **Research through Design (RtD)** studies require consideration of the point at which new consent from participants is warranted. We remained well within our existing ethical approval, creating and discussing an alternative dashboard inspired by previous conversations with participants who were already familiar with the inferences possible via energy monitoring. Yet a design involving more comprehensive data analysis or deployed amongst less familiar or less energy literate participants may have overstepped this mark. Hence, we stress that Household Wattch was carefully tailored to its audience and may not be suitable for a new or unfamiliar eco-feedback cohort. If adapting the method, careful consideration is necessary to ensure provotypes are contextually and ethically appropriate. More generally, researchers should be cognizant of the potentially grey area of if/when to obtain re-consent in longitudinal RtD work, given it is seldom possible at the outset of a project (or even at the outset of an activity within a project) to anticipate precisely what themes may be raised.

5.3 Limitations and Future Work

The limited sample size owes to our decision to maximise the authenticity and legitimacy of Household Wattch by utilising participants' own energy use data. This meant engaging existing eco-feedback trial participants who had consented to sharing their energy data with us rather than seeking new participants. The existing relationships, including the trust and rapport built between the researchers and the participants may introduce bias, such as envisaging Household Wattch more positively. Yet equally, we argue that this level of existing trust and rapport is vital when introducing and discussing potentially sensitive issues, such as inference from energy use data. We would be hesitant to run a similar study with a completely unknown group of participants. Further

⁶<https://www.merriam-webster.com/dictionary/provoke>

work is warranted into how responses may differ in a similar study run by energy utilities rather than a university, given the progressively decreased trust placed in Australia's energy sector [27].

The sample being comprised entirely of co-habiting couples with/without children is a strength of the study, given the small sample size precludes the ability to compare groups. Yet equally, the findings do not represent household types other than families, e.g., shared houses, boarding houses or workers' accommodation. Perspectives of other family members would have been valuable, even if they were otherwise disengaged with household energy use. Yet equally, *requiring* rather than *inviting* other householders to participate would have almost certainly further reduced our already limited sample size and lost valuable perspectives. Irrespective of our sampling, the study highlights a range of ethical considerations resultant from the interpersonal nature of energy use data. In particular, there is a need to further investigate the effect of power dynamics between eco-feedback lead users and others within a household [44] and the implications for surveillance.

6 Conclusion

Household Wattch provides a novel user-centred perspective into a near future of more ubiquitous high-granularity household energy use data. It highlights the opportunities and risks that such a future brings and discusses how design might best advocate for users in this future. The article demonstrates how provotypes can engage households in complex issues such as attribution, commodification and surveillance through their energy use data. We identify necessary human-centred design supports to enable a more consensual energy future and detail priorities for design, including: (1) exploring alternative avenues for eco-feedback beyond energy saving, (2) assisting households in navigating the complexities of consent when sharing interpersonal energy use data beyond the home and (3) identifying and avoiding opportunities for anti-social surveillance through energy use data. The complexity of the issues raised highlights the need for much future work in this space towards more consensual and positive energy futures.

Appendix

A Household Wattch Metrics and Scoring

The interface design and choice of metrics were curated by the research team comprised of two HCI designers and a data analyst over multiple iterations. The data analyst explored the data in detail using R and Python, with a view to understand the depth and variety of inferences possible with the available energy use data: i.e., 10-second data capture frequency with per-circuit submetering on each household. Then the HCI designers prototyped different methods for scoring and different formats of data presentation, culminating in the present provotype. The components (*Home Insurance Risk*, *Household Health*, *Sustainability Index*, and *Appliance Health Check*) and their constituents are presented below in Table 2; the product of ongoing collaboration between the data analyst and HCI designers.

Table 2. Details of Each of the Four Household Wattch Metrics, Including Analysis Method, Scoring Protocol for Each and Justifications for Their Inclusion in the App

Inference	Analysis method	Justification	Scoring
<i>Household health</i>			
Sleep duration	Analysis of lighting circuit power to approximate [bedtime] and [wake time].	Sleep duration and disturbance can affect mental health status and next-day work performance (e.g., [58, 59]).	Sleep duration: >7 hours = 0 5–7 hours = –1 < 5 hours = –2
	Sleep duration = hours between bedtime and wake up time.	The upper limit of 100 W means the algorithm does not pick up external sensor lights, which may be triggered by outside events or wildlife.	Sleep disturbance: 0–1 events per night = 0 1–2 events per night = –1 > 2 events per night = –2
Sleep disturbance	Between [household bedtime] and [household wake time], define lighting circuit peaks of 8 W–100 W of less than 30 minutes duration as sleep disturbance events.		
Microwave and stove top use	Microwave: (1) Visual determination of which circuit corresponds to kitchen sockets. (2) Once determined, on this circuit, classify peaks between 600 and 1,200 W for up to 5 minutes' duration as microwave events.	Estimation of eating habits, e.g., low stove use + high microwave use may indicate tendency for microwave dinners. Low stove use + low microwave use may indicate busy lifestyle and disposable income for restaurant or takeout.	Low stove + low microwave = –1 Low stove + high microwave = –1 High stove + low microwave = 0
	Stovetop/oven: Peaks of 500–5000 W being used for 1–120 minutes were flagged as stove/oven events.	High stove use + low microwave use may indicate healthier eating, cooking from fresh.	High stove + high microwave = 0
<i>Home insurance risk</i>			
Oven left on	For households with a dedicated circuit for the oven, any peaks of > 800 W for a period of > 2.5 hours were counted.	Oven-left-on events are associated with dementia and forgetfulness [39].	Each >2.5 hour oven on = –2
Occupancy	Consumption peaks recorded throughout the day, consistent activity on power points and/or lighting and/or air conditioning circuit on.	Household occupancy patterns provide a proxy for social contact, sedentary behaviour, estimation of occupation type, e.g., shift worker vs 9–5 worker.	High occupancy = 1 Medium occupancy = 0 Low occupancy = –1
Gas appliance ownership	Check participant demographic data for ownership of one or more gas appliance.	Gas appliance ownership may indicate higher likelihood of fire or explosion relative to houses without any gas connection fitted.	Owns gas appliances = –2 Else = 0
Age risk (< 25 or > 80 years)	Check participant demographic data for age of adult participants.	Houses with all adult participants < 25 or > 80 years old may indicate increased likelihood of damage to house or fire risk through misadventure or carelessness.	< 25 or > 80 years old = –2 Else = 0
<i>Sustainability index</i>			
Consumption / Grid import	Derived from consumption (sum of all circuits) and net (consumption minus production) for households with solar PV.	High total electricity consumption (non-solar PV owners) or high total grid import (solar PV owners) indicates less sustainable than those with lower consumption/grid import	Total consumption values normalised (0–1) across all households who do not own solar PV Total grid import values normalised (0–1) across all households who do own solar PV.
Judicious heating/cooling	Using the dedicated aircon circuit, identify start end and length of continuous activity >20 W (indicating operation rather than standby power). Then: Regression to correlate air conditioning use to publicly available local weather data (Australian Bureau of Meteorology) to determine temperature thresholds for AC operation, e.g., likelihood of a given householder turning on AC at a given temperature.	Temperature sensitivity indicates comfort preferences and acts as a potential indicator for health. For example obesity is correlated to feelings of warm discomfort [85]. Ill-health drives higher heating demand in cooler environments [67].	High temperature sensitivity = –1 or –2 depending on severity. Neutral temperature sensitivity = 0.

Scores were computed for each metric for each household. The scores for each metric were then normalised across all 16 households in the sample, meaning each household's score for each metric could be represented as a value between 0 and 100 and placed on a continuum showing its relation to the "Neighbourhood Average" (refer Figure 1 in the article).

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Authors' Statement

This article represents original work by the authors that is not under consideration anywhere else or in any other format. None of the authors report a conflict of interest. The article relates to prior work of the lead author, who has investigated householders' experiences with energy use feedback before (e.g., the article "An Eco-feedback Joins the Family") and touched on privacy in this domain (e.g., the article "Neighbourhood Wattch: Using Speculative Design to Explore Values Around Curtailment and Consent in Energy Data"). However, the sample group for this article is entirely different, and the focus is on energy data at the household scale, not on comparison with others, social normative comparison or on incentives for voluntary curtailment.

Conflict of Interest Statement

No authors report a conflict of interest of any nature with regard to the study or participants.

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